

EXTENSION 1

Stage 6

Combinatorics (Ext1), A1 Working with Combinatorics (Y11)

Binomial Expansion (Ext1)

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Exam Equivalent Time: 79.5 minutes (based on allocation of 1.5 minutes per mark)

A1 Combinatorics



*SmarterMaths analytics based on the average contribution to Extension 1 exams since 2020.

HISTORICAL CONTRIBUTION

- Combinatorics has contributed a healthy average of 9.3% per new syllabus Ext1 exam since it was introduced in 2020.
- This topic has been split into two sub-categories for analysis purposes which are: 1-Permutations and Combinations (4.3%) and 2-Binomial Expansion (5.0%).
- This analysis looks at Binomial Expansion.

HSC ANALYSIS - What to expect and common pitfalls

- Binomial Expansion (5.0%) has been tested in the longer answer section of each new syllabus Ext1 exam to date.
- This topic area can be examined over a very wide range of difficulties.
- 2020 Ext1 14a tested this topic area with an extremely challenging 8-mark question that reframed the possible level of difficulty of binomial proofs that may appear. An important revision question.
- The 2021 exam, in contrast, served up a band 3 softball that was a gift to students.
- A common question type, last examined in 2022, requires the calculation of the constant/independent term or a specific coefficient in a given binomial expansion. We note that the NESA topic guidance states that "greatest coefficient" questions are no longer examinable and have been removed from the database.
- These types of questions are generally well answered, although significant problems were caused in 2012 Ext1 11f and 2005 Ext1 2b and should be carefully reviewed.
- Combinatorial proofs that receive a specific mention in the syllabus are covered in the database.

Questions

1. Combinatorics, EXT1 A1 2014 HSC 3 MC

What is the constant term in the binomial expansion of $\left(2x - \frac{5}{x^3}\right)^{12}$?

- A. $\binom{12}{3}2^95^3$
- B. $\binom{12}{9}2^35^9$
- C. $-\binom{12}{3}2^95^3$
- D. $-\binom{12}{9}2^35^9$

2. Combinatorics, EXT1 A1 2017 HSC 9 MC

When expanded, which expression has a non-zero constant term?

- A. $\left(x + \frac{1}{x^2}\right)^7$
- B. $\left(x^2 + \frac{1}{x^3}\right)^7$
- C. $\left(x^3 + \frac{1}{x^4}\right)^7$
- D. $\left(x^4 + \frac{1}{x^5}\right)^7$

3. Combinatorics, EXT1' A1 2016 HSC 6 MC

Let $p(x) = 1 + x + x^2 + x^3 + \dots + x^{12}$.

What is the coefficient of x^8 in the expansion of $p(x + 1)$?

- A. 1
- B. 495
- C. 715
- D. 1287

4. Combinatorics, EXT1 A1 EQ-Bank 8

Show ${}^nC_k = {}^nC_{n-k}$. (1 mark)

5. Combinatorics, EXT1 A1 SM-Bank 9

$$\left(2 - \sqrt{3}\right)^5 = a + b\sqrt{3}.$$

Evaluate a and b using a binomial expansion. (2 marks)

6. Combinatorics, EXT1 A1 2021 HSC 11b

Expand and simplify $(2a - b)^4$. (2 marks)

7. Combinatorics, EXT1 A1 2015 HSC 13b

Consider the binomial expansion

$$\left(2x + \frac{1}{3x}\right)^{18} = a_0x^{18} + a_1x^{16} + a_2x^{14} + \dots$$

where $a_0, a_1, a_2, \dots$ are constants.

- i. Find an expression for a_2 . (2 marks)
- ii. Find an expression for the term independent of x . (2 marks)

8. Combinatorics, EXT1 A1 2008 HSC 1d

Find an expression for the coefficient of x^8y^4 in the expansion of $(2x + 3y)^{12}$. (2 marks)

9. Combinatorics, EXT1 A1 2011 HSC 2c

Find an expression for the coefficient of x^2 in the expansion of $\left(3x - \frac{4}{x}\right)^8$. (2 marks)

10. Combinatorics, EXT1 A1 2019 13b

In the expansion of $(5x + 2)^{20}$, the coefficients of x^r and x^{r+1} are equal.
What is the value of r ? (3 marks)

11. Combinatorics, EXT1 A1 2022 HSC 11c

Find the coefficients of x^2 and x^3 in the expansion of $\left(1 - \frac{x}{2}\right)^8$. (2 marks)

12. Combinatorics, EXT1 A1 2023 HSC 12d

It is known that ${}^nC_r = {}^{n-1}C_{r-1} + {}^{n-1}C_r$ for all integers such that $1 \leq r \leq n - 1$. (Do NOT prove this.)

Find ONE possible set of values for p and q such that

$${}^{2022}C_{80} + {}^{2022}C_{81} + {}^{2023}C_{1943} = {}^pC_q$$
 (2 marks)

13. Combinatorics, EXT1 A1 EQ-Bank 15

$$\left(\sqrt{5} - 2\right)^5 = x + y\sqrt{5}$$

Find the values of x and y using binomial expansion. (2 marks)

14. Combinatorics, EXT1 A1 SM-Bank 1

$$\left(1 - 2\sqrt{2}\right)^6 = x + y\sqrt{2}$$

Evaluate the value of x and y . (2 marks)

15. Combinatorics, EXT1 A1 EQ-Bank 3

Find the coefficient of x^4 in the expansion of $\left(x^2 - \frac{3}{x}\right)^5$. (2 marks)

16. Combinatorics, EXT1 A1 EQ-Bank 7

Show ${}^nC_k = {}^{n-1}C_{k-1} + {}^{n-1}C_k$. (2 marks)

17. Combinatorics, EXT1 A1 EQ-Bank 10

By using the fact that $(1 + x)^{11} = (1 + x)^3(1 + x)^8$, show that

$$\binom{11}{5} = \binom{8}{5} + \binom{3}{1}\binom{8}{4} + \binom{3}{2}\binom{8}{3} + \binom{8}{2}.$$
 (3 marks)

18. Combinatorics, EXT1 A1 2019 MET1 8

A fair standard die is rolled 50 times. Let W be a random variable with binomial distribution that represents the number of times the face with a six on it appears uppermost.

- a. Write down the expression for $P(W = k)$, where $k \in \{0, 1, 2, \dots, 50\}$. (1 mark)
- b. Show that $\frac{P(W = k + 1)}{P(W = k)} = \frac{50 - k}{5(k + 1)}$. (2 marks)

19. Combinatorics, EXT1 A1 2005 HSC 2b

Use the binomial theorem to find the term independent of x in the expansion of

$$\left(2x - \frac{1}{x^2}\right)^{12}. \quad (3 \text{ marks})$$

20. Combinatorics, EXT1 A1 SM-Bank 2

Using $(1+x)^4(1+x)^9 = (1+x)^{13}$

show that

$${}^9C_4 + {}^4C_1 {}^9C_3 + {}^4C_2 {}^9C_2 + {}^4C_3 {}^9C_1 + 1 = {}^{13}C_4 \quad (2 \text{ marks})$$

21. Combinatorics, EXT1 A1 2012 HSC 11f

i. Use the binomial theorem to find an expression for the constant term in the expansion of

$$\left(2x^3 - \frac{1}{x}\right)^{12}. \quad (2 \text{ marks})$$

ii. For what values of n does $\left(2x^3 - \frac{1}{x}\right)^n$ have a non-zero constant term? (1 mark)

22. Combinatorics, EXT1 A1 2020 HSC 14a

i. Use the identity $(1+x)^{2n} = (1+x)^n(1+x)^n$ to show that

$$\binom{2n}{n} = \binom{n}{0}^2 + \binom{n}{1}^2 + \dots + \binom{n}{n}^2,$$

where n is a positive integer. (2 marks)

ii. A club has $2n$ members, with n women and n men.

A group consisting of an even number $(0, 2, 4, \dots, 2n)$ of members is chosen, with the number of men equal to the number of women.

Show, giving reasons, that the number of ways to do this is $\binom{2n}{n}$. (2 marks)

iii. From the group chosen in part (ii), one of the men and one of the women are selected as leaders.

Show, giving reasons, that the number of ways to choose the even number of people and then the leaders is

$$1^2 \binom{n}{1}^2 + 2^2 \binom{n}{2}^2 + \dots + n^2 \binom{n}{n}^2. \quad (2 \text{ marks})$$

iv. The process is now reversed so that the leaders, one man and one woman, are chosen first. The rest of the group is then selected, still made up of an equal number of women and men.

By considering this reversed process and using part (ii), find a simple expression for the sum in part (iii). (2 marks)

Worked Solutions

1. Combinatorics, EXT1 A1 2014 HSC 3 MC

General term

$$\begin{aligned} T_k &= \binom{12}{k} (2x)^{12-k} \cdot (-1)^k \cdot (5x^{-3})^k \\ &= \binom{12}{k} 2^{12-k} \cdot x^{12-k} \cdot (-1)^k \cdot 5^k \cdot x^{-3k} \\ &= \binom{12}{k} (-1)^k \cdot 2^{12-k} \cdot 5^k \cdot x^{12-4k} \end{aligned}$$

Constant term when

$$\begin{aligned} 12-4k &= 0 \\ k &= 3 \end{aligned}$$

$\therefore$ Constant term

$$\begin{aligned} &= \binom{12}{3} (-1)^3 \cdot 2^9 \cdot 5^3 \\ &= -\binom{12}{3} \cdot 2^9 \cdot 5^3 \end{aligned}$$

$$\Rightarrow C$$

2. Combinatorics, EXT1 A1 2017 HSC 9 MC

Consider the general term for option A:

$$\begin{aligned} T_k &= {}^7C_k \cdot x^{7-k} \cdot x^{-2k} \\ &= {}^7C_k \cdot x^{7-3k} \end{aligned}$$

Non zero constant term occurs when

$$\begin{aligned} 7-3k &= 0 \\ k &= \frac{7}{3} \Rightarrow \text{no terms exists } (k \text{ not integer}) \end{aligned}$$

Consider option C:

$$\begin{aligned} T_k &= {}^7C_k \cdot x^{3(7-k)} \cdot x^{-4k} \\ &= {}^7C_k \cdot x^{21-7k} \end{aligned}$$

$$\begin{aligned} 21-7k &= 0 \\ k &= 3 \end{aligned}$$

$\therefore$ Non-zero constant term exists

since k is an integer

$$\Rightarrow C$$

Worked Solutions

3. Combinatorics, EXT1' A1 2016 HSC 6 MC

$$\begin{aligned} p(x+1) &= 1 + (x+1) + (x+1)^2 + \dots + (x+1)^{12} \\ \text{Coefficient of } x^8 &= {}^{12}C_8 + {}^{11}C_8 + {}^{10}C_8 + {}^9C_8 + {}^8C_8 \\ &= 715 \\ &\Rightarrow C \end{aligned}$$

4. Combinatorics, EXT1 A1 EQ-Bank 8

$$\begin{aligned} {}^nC_k &= \frac{n!}{(n-k)!k!} \\ {}^nC_{n-k} &= \frac{n!}{(n-(n-k))!(n-k)!} \\ &= \frac{n!}{k!(n-k)!} \\ &= {}^nC_k \end{aligned}$$

5. Combinatorics, EXT1 A1 SM-Bank 9

$$\begin{aligned} (2-\sqrt{3})^5 &= {}^5C_0 \cdot 2^5 + {}^5C_1 \cdot 2^4(-\sqrt{3}) + {}^5C_2 \cdot 2^3(-\sqrt{3})^2 + {}^5C_3 \cdot 2^2(-\sqrt{3})^3 \\ &\quad + {}^5C_4 \cdot 2(-\sqrt{3})^4 + {}^5C_5(-\sqrt{3})^5 \\ &= 32-80\sqrt{3}+240-120\sqrt{3}+90-9\sqrt{3} \\ &= 362-209\sqrt{3} \end{aligned}$$

$$\therefore a = 362, b = -209$$

6. Combinatorics, EXT1 A1 2021 HSC 11b

$$\begin{aligned} (2a-b)^4 &= (2a)^4 + {}^4C_1(2a)^3(-b) + {}^4C_2(2a)^2(-b)^2 + {}^4C_3(2a)(-b)^3 + (-b)^4 \\ &= 16a^4 - 4 \cdot 8a^3b + 6 \cdot 4a^2b^2 - 4 \cdot 2ab^3 + b^4 \\ &= 16a^4 - 32a^3b + 24a^2b^2 - 8ab^3 + b^4 \end{aligned}$$

7. Combinatorics, EXT1 A1 2015 HSC 13b

i. Need co-efficient of x^{14}

$$\text{General term of } \left(2x + \frac{1}{3x}\right)^{18}$$

$$\begin{aligned} T_k &= {}^{18}C_k (2x)^{18-k} \cdot \left(\frac{1}{3x}\right)^k \\ &= {}^{18}C_k \cdot 2^{18-k} \cdot x^{18-k} \cdot 3^{-k} \cdot x^{-k} \\ &= {}^{18}C_k \cdot 2^{18-k} \cdot 3^{-k} \cdot x^{18-2k} \end{aligned}$$

a_2 occurs when:

$$\begin{aligned} 18 - 2k &= 14 \\ 2k &= 4 \\ k &= 2 \end{aligned}$$

$$\begin{aligned} \therefore a_2 &= {}^{18}C_2 \cdot 2^{18-2} \cdot 3^{-2} \\ &= \frac{{}^{18}C_2 \cdot 2^{16}}{3^2} \end{aligned}$$

ii. Independent term occurs when:

$$\begin{aligned} 18 - 2k &= 0 \\ 2k &= 18 \\ k &= 9 \end{aligned}$$

$$\begin{aligned} \therefore \text{Independent term} &= {}^{18}C_9 \cdot 2^{18-9} \cdot 3^{-9} \\ &= \frac{{}^{18}C_9 \cdot 2^9}{3^9} \end{aligned}$$

8. Combinatorics, EXT1 A1 2008 HSC 1d

Find co-efficient of $x^8 y^4$:

$$T_k = \text{General term of } (2x + 3y)^{12}$$

$$\begin{aligned} T_k &= \binom{12}{k} (2x)^{12-k} \cdot (3y)^k \\ &= \binom{12}{k} \cdot 2^{12-k} \cdot 3^k \cdot x^{12-k} \cdot y^k \end{aligned}$$

$x^8 y^4$ occurs when $k = 4$

$$T_4 = \binom{12}{4} \cdot 2^{12-4} \cdot 3^4 \cdot x^8 y^4$$

$$\begin{aligned} \therefore \text{Co-efficient of } x^8 y^4 &= \binom{12}{4} \cdot 2^8 \cdot 3^4 \\ &= 10\,264\,320 \end{aligned}$$

MARKER'S COMMENT: More errors were made by students who used T_{k+1} as the general term rather than T_k (both are possible). The Worked Solution uses the more successful approach.

9. Combinatorics, EXT1 A1 2011 HSC 2c

$$\left(3x - \frac{4}{x}\right)^8 = \sum_{k=0}^8 {}^8C_k \cdot (3x)^{8-k} \cdot \left(-\frac{4}{x}\right)^k$$

$$\begin{aligned} \text{General term} &= {}^8C_k \cdot 3^{8-k} \cdot x^{8-k} \cdot (-1)^k \cdot 4^k \cdot x^{-k} \\ &= {}^8C_k \cdot (-1)^k \cdot 3^{8-k} \cdot 4^k \cdot x^{8-2k} \end{aligned}$$

Co-efficient of $x^2 = 2$ occurs when,

$$\begin{aligned} 8 - 2k &= 2 \\ 2k &= 6 \\ k &= 3 \end{aligned}$$

$$\begin{aligned} \therefore \text{Co-efficient of } x^2 &= {}^8C_3 \cdot (-1)^3 \cdot 3^5 \cdot 4^3 \\ &= -56 \times 243 \times 64 \\ &= -870\,912 \end{aligned}$$

10. Combinatorics, EXT1 A1 2019 13b

$$(5x + 2)^{20}$$

General term: $(2 + 5x)^{20}$

$$T_r = {}^{20}C_r \cdot 2^{20-r} \cdot (5x)^r$$

$$= {}^{20}C_r \cdot 2^{20-r} \cdot 5^r \cdot x^r$$

$$T_{r+1} = {}^{20}C_{r+1} \cdot 2^{19-r} \cdot 5^{r+1} \cdot x^{r+1}$$

Equating co-efficients:

$$\frac{20!}{r!(20-r)!} \cdot 2^{20-r} \cdot 5^r = \frac{20!}{(r+1)!(19-r)!} \cdot 2^{19-r} \cdot 5^{r+1}$$

$$\frac{2}{20-r} = \frac{5}{r+1}$$

$$2r + 2 = 100 - 5r$$

$$7r = 98$$

$$r = 14$$

11. Combinatorics, EXT1 A1 2022 HSC 11c

General Term: $\left(1 - \frac{x}{2}\right)^8$

$$T_k = \binom{8}{k} (1)^{8-k} \left(-\frac{x}{2}\right)^k$$

$$= \binom{8}{k} \left(-\frac{1}{2}\right)^k x^k$$

Coefficient of $x^2 = \binom{8}{2} \left(-\frac{1}{2}\right)^2 = 7$

Coefficient of $x^3 = \binom{8}{3} \left(-\frac{1}{2}\right)^3 = -7$

COMMENT: Arithmetic becomes easier by expanding $(2 + 5x)^{20}$.

12. Combinatorics, EXT1 A1 2023 HSC 12d

$${}^{2022}C_{80} + {}^{2022}C_{81} + {}^{2023}C_{1943} = {}^pC_q$$

Using the known relationship:

$${}^{2022}C_{80} + {}^{2022}C_{81} = {}^{2023}C_{81} \dots (1)$$

Also, since ${}^nC_r = {}^nC_{n-r}$

$${}^{2023}C_{1943} = {}^{2023}C_{2023-1943} = {}^{2023}C_{80} \dots (2)$$

$$\begin{aligned} {}^{2022}C_{80} + {}^{2022}C_{81} + {}^{2023}C_{1943} &= {}^{2023}C_{81} + {}^{2023}C_{1943} \text{ (see (1) above)} \\ &= {}^{2023}C_{81} + {}^{2023}C_{80} \text{ (see (2) above)} \\ &= {}^{2024}C_{81} \end{aligned}$$

$$\therefore p = 2024, q = 81$$

13. Combinatorics, EXT1 A1 EQ-Bank 15

$$\begin{aligned} (\sqrt{5} - 2)^5 &= {}^5C_0 \cdot (\sqrt{5})^5 + {}^5C_1 (\sqrt{5})^4 \cdot (-2) + {}^5C_2 (\sqrt{5})^3 \cdot (-2)^2 \\ &\quad + {}^5C_3 (\sqrt{5})^2 \cdot (-2)^3 + {}^5C_4 (\sqrt{5}) \cdot (-2)^4 + {}^5C_5 (-2)^5 \\ &= 25\sqrt{5} - 250 + 200\sqrt{5} - 400 + 80\sqrt{5} - 32 \\ &= 305\sqrt{5} - 682 \end{aligned}$$

$$\therefore x = -682, y = 305$$

14. Combinatorics, EXT1 A1 SM-Bank 1

Using the binomial expansion:

$$\begin{aligned} (1 - 2\sqrt{2})^6 &= {}^6C_0 + {}^6C_1 (-2\sqrt{2}) + {}^6C_2 (-2\sqrt{2})^2 + {}^6C_3 (-2\sqrt{2})^3 + {}^6C_4 (-2\sqrt{2})^4 \\ &\quad + {}^6C_5 (-2\sqrt{2})^5 + {}^6C_6 (-2\sqrt{2})^6 \end{aligned}$$

$$\begin{aligned} &= 1 - 12\sqrt{2} + 120 - 320\sqrt{2} + 960 - 768\sqrt{2} + 512 \\ &= 1593 - 1100\sqrt{2} \end{aligned}$$

$$\therefore x = 1593, y = -1100$$

15. Combinatorics, EXT1 A1 EQ-Bank 3

General term:

$$\begin{aligned} T_k &= {}^5C_k (x^2)^{5-k} \cdot \left(-\frac{3}{x}\right)^k \\ &= {}^5C_k \cdot x^{10-2k} (-3)^k \cdot x^{-k} \\ &= {}^5C_k \cdot x^{10-3k} \cdot (-3)^k \end{aligned}$$

Coefficient of x^4 occurs when

$$\begin{aligned} 10-3k &= 4 \\ 3k &= 6 \\ k &= 2 \end{aligned}$$

$$\begin{aligned} \therefore \text{Coefficient of } x^4 &= {}^5C_2 \cdot (-3)^2 \\ &= 90 \end{aligned}$$

16. Combinatorics, EXT1 A1 EQ-Bank 7

$$\text{LHS} = \frac{n!}{(n-k)!k!}$$

$$\begin{aligned} \text{RHS} &= \frac{(n-1)!}{(n-1-(k-1))!(k-1)!} + \frac{(n-1)!}{(n-1-k)!k!} \\ &= \frac{(n-1)!k}{(n-k)!(k-1)!k} + \frac{(n-1)!(n-k)}{(n-k-1)!(n-k)k!} \\ &= \frac{(n-1)!k}{(n-k)!k!} + \frac{(n-1)!(n-k)}{(n-k)!k!} \\ &= \frac{(n-1)!(k+n-k)}{(n-k)!k!} \\ &= \frac{n!}{(n-k)!k!} \\ &= \text{LHS} \end{aligned}$$

17. Combinatorics, EXT1 A1 EQ-Bank 10

General term of $(1+x)^{11}$:

$$\begin{aligned} T_k &= {}^{11}C_k \cdot 1^{11-k} \cdot x^k \\ \Rightarrow {}^{11}C_5 &\text{ is the co-efficient of } x^5 \end{aligned}$$

$$\begin{aligned} (1+x)^3 &= {}^3C_0 + {}^3C_1x + {}^3C_2x^2 + {}^3C_3x^3 \\ (1+x)^8 &= {}^8C_0 + {}^8C_1x + {}^8C_2x^2 + {}^8C_3x^3 + {}^8C_4x^4 + {}^8C_5x^5 + \dots \end{aligned}$$

$$\begin{aligned} \therefore \text{Coefficient of } x^5 \text{ in } (1+x)^3(1+x)^8 &= {}^3C_0 \cdot {}^8C_5 + {}^3C_1 \cdot {}^8C_4 + {}^3C_2 \cdot {}^8C_3 + {}^3C_3 \cdot {}^8C_2 \\ &= {}^8C_5 + {}^3C_1 \cdot {}^8C_4 + {}^3C_2 \cdot {}^8C_3 + {}^8C_2 \end{aligned}$$

Equating coefficients:

$$\binom{11}{5} = \binom{8}{5} + \binom{3}{1}\binom{8}{4} + \binom{3}{2}\binom{8}{3} + \binom{8}{2}$$

18. Combinatorics, EXT1 A1 2019 MET1 8

$$\text{a. } P(W=k) = {}^{50}C_k \cdot \left(\frac{1}{6}\right)^k \cdot \left(\frac{5}{6}\right)^{50-k}$$

$$\begin{aligned} \text{b. } \frac{P(W=k+1)}{P(W=k)} &= \frac{{}^{50}C_{k+1} \cdot \left(\frac{1}{6}\right)^{k+1} \cdot \left(\frac{5}{6}\right)^{49-k}}{{}^{50}C_k \cdot \left(\frac{1}{6}\right)^k \cdot \left(\frac{5}{6}\right)^{50-k}} \\ &= \frac{\frac{50!}{(49-k)!(k+1)!} \cdot \left(\frac{1}{6}\right)}{\frac{50!}{(50-k)!k!} \cdot \left(\frac{5}{6}\right)} \\ &= \frac{(50-k)!k!}{5(49-k)!(k+1)!} \\ &= \frac{50-k}{5(k+1)} \end{aligned}$$

19. Combinatorics, EXT1 A1 2005 HSC 2b

$$T_k = \text{General term of } \left(2x - \frac{1}{x^2}\right)^{12}$$

$$T_k = \binom{12}{k} (2x)^{12-k} \cdot (-1)^k \cdot (x^{-2})^k$$

$$= \binom{12}{k} \cdot 2^{12-k} \cdot x^{12-k} \cdot (-1)^k \cdot x^{-2k}$$

$$= \binom{12}{k} \cdot 2^{12-k} \cdot (-1)^k \cdot x^{12-3k}$$

Independent term occurs when

$$x^{12-3k} = x^0$$

$$12-3k = 0$$

$$k = 4$$

MARKER'S COMMENT: The general term formula was well known, but many could not apply it to this question.

∴ Independent term is

$$\binom{12}{4} \cdot 2^8 \cdot (-1)^4 = 126 \cdot 720$$

20. Combinatorics, EXT1 A1 SM-Bank 2

Expanding $(1+x)^{13}$:

$$T_k = {}^{13}C_k \cdot 1^{13-k} \cdot x^k$$

$$\Rightarrow {}^{13}C_4 \text{ is coefficient of } x^4$$

$$(1+x)^4 = {}^4C_0 + {}^4C_1x + {}^4C_2x^2 + {}^4C_3x^3 + {}^4C_4x^4$$

$$(1+x)^9 = {}^9C_0 + {}^9C_1x + {}^9C_2x^2 + {}^9C_3x^3 + {}^9C_4x^4 + \dots$$

$$\therefore \text{Coefficient of } x^4 \text{ in } (1+x)^4(1+x)^9$$

$$= {}^4C_0 \cdot {}^9C_4 + {}^4C_1 \cdot {}^9C_3 + {}^4C_2 \cdot {}^9C_2 + {}^4C_3 \cdot {}^9C_1 + {}^4C_4 \cdot {}^9C_0$$

$$= {}^9C_4 + {}^4C_1 \cdot {}^9C_3 + {}^4C_2 \cdot {}^9C_2 + {}^4C_3 \cdot {}^9C_1 + 1$$

$$= {}^{13}C_4 \dots \text{as required}$$

21. Combinatorics, EXT1 A1 2012 HSC 11f

i. General term

$${}^{12}C_k \cdot (2x^3)^{12-k} \left(-\frac{1}{x}\right)^k$$

$$= {}^{12}C_k \cdot (-1)^k \cdot 2^{12-k} \cdot x^{36-3k} \cdot x^{-k}$$

$$= {}^{12}C_k \cdot (-1)^k \cdot 2^{12-k} \cdot x^{36-4k}$$

Constant term occurs when

$$36-4k = 0$$

$$k = 9$$

$$\therefore \text{Constant term} = {}^{12}C_9 \cdot (-1)^9 \cdot 2^3$$

$$= -\frac{12!}{3!9!} \times 8$$

$$= -1760$$

ii. General term of $\left(2x^3 - \frac{1}{x}\right)^n$

$${}^nC_k \cdot (2x^3)^{n-k} \left(-\frac{1}{x}\right)^k$$

$$= {}^nC_k \cdot 2^{n-k} \cdot x^{3n-3k} \cdot (-1)^k \cdot x^{-k}$$

$$= {}^nC_k \cdot (-1)^k \cdot 2^{n-k} \cdot x^{3n-4k}$$

◆◆◆ Mean mark 16%.

Constant term when $3n-4k = 0$.

$$\text{i.e. } k = \frac{3}{4}n$$

Since n and k must be integers, n must be a multiple of 4.

i. Expand $(1 + x)^{2n}$:

$${}^{2n}C_0 + {}^{2n}C_1x^2 + \dots + {}^{2n}C_nx^n + \dots {}^{2n}C_{2n}x^{2n}$$

$\Rightarrow$ Coefficient of $x^n = {}^{2n}C_n$

Expand $(1 + x)^n(1 + x)^n$:

$$[{}^nC_0 + {}^nC_1x + \dots + {}^nC_nx^n][{}^nC_0 + {}^nC_1x + \dots + {}^nC_nx^n]$$

$\Rightarrow$ Coefficient of x^n

$$= {}^nC_0 \cdot {}^nC_n + {}^nC_1 \cdot {}^nC_{n-1} + \dots + {}^nC_{n-1} \cdot {}^nC_1 + {}^nC_n \cdot {}^nC_0$$

$$= ({}^nC_0)^2 + ({}^nC_1)^2 + \dots + ({}^nC_{n-1})^2 + ({}^nC_n)^2 \quad ({}^nC_k = {}^nC_{n-k})$$

Equating coefficients:

$${}^{2n}C_n = ({}^nC_0)^2 + ({}^nC_1)^2 + \dots + ({}^nC_n)^2$$

♦♦ Mean mark part (i) 26%.

ii. Number of men = Number of women ($M = W$)

If $M = W = 0$: Ways = ${}^nC_0 \cdot {}^nC_0 = ({}^nC_0)^2$

If $M = W = 1$: Ways = ${}^nC_1 \cdot {}^nC_1 = ({}^nC_1)^2$

$\vdots$

If $M = W = n$: Ways = ${}^nC_n \cdot {}^nC_n = ({}^nC_n)^2$

$\therefore$ Total combinations

$$= ({}^nC_0)^2 + ({}^nC_1)^2 + \dots + ({}^nC_n)^2$$

$$= {}^{2n}C_n \quad (\text{from part (i)})$$

♦♦ Mean mark part (ii) 23%.

iii. Let M_L = possible male leaders

Let W_L = possible female leaders

If $M = W = 0 \Rightarrow$ no leaders

If $M = W = 1$: Ways = ${}^nC_1 \times M_L \times {}^nC_1 \times W_L = 1^2 ({}^nC_1)^2$

If $M = W = 2$: Ways = ${}^nC_2 \times 2 \times {}^nC_2 \times 2 = 2^2 ({}^nC_2)^2$

$\vdots$

If $M = W = n$: Ways = ${}^nC_n \times n \times {}^nC_n \times n = n^2 ({}^nC_n)^2$

$\therefore$ Total combinations

$$= 1^2 ({}^nC_1)^2 + 2^2 ({}^nC_2)^2 + \dots + n^2 ({}^nC_n)^2$$

♦♦ Mean mark part (iii) 26%.

iv. If $M = W = 1$: Ways = $M_L \times {}^{n-1}C_0 \times W_L \times {}^{n-1}C_0$

$$= n \times {}^{n-1}C_0 \times n \times {}^{n-1}C_0$$

$$= n^2 ({}^{n-1}C_0)^2$$

♦♦♦ Mean mark part (iv) 16%.

If $M = W = 2$: Ways = $n \times {}^{n-1}C_1 \times n \times {}^{n-1}C_1$

$$= n^2 ({}^{n-1}C_1)^2$$

$\vdots$

If $M = W = n$: Ways = $n \times {}^{n-1}C_{n-1} \times n \times {}^{n-1}C_{n-1}$

$$= n^2 ({}^{n-1}C_{n-1})^2$$

$\therefore$ Total combinations

$$= n^2 ({}^{n-1}C_0)^2 + n^2 ({}^{n-1}C_1)^2 + \dots + n^2 ({}^{n-1}C_{n-1})^2$$

$$= n^2 \times {}^{2(n-1)}C_{n-1} \quad (\text{using part (i)})$$

$$= n^2 \times {}^{2n-2}C_{n-1}$$